Can LLMs Predict Human Behavior? A Measure of their Pretrained Knowledge

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Introduction

Large language models (LLMs) are large-scale neural networks trained on vast corpora of internet text

A striking feature of LLMs:

- mimic human behaviors across a wide range of domains
- because of extensive **pretraining** on human behaviors
- e.g., LLMs reproduce human behavioral biases (anchoring, framing, loss aversion) (Jones & Steinhardt 22, Horton 23)

Thus increasingly used to **predict** human behaviors, e.g.,

- predicting outcomes of economic experiments (Aher et al 23, Hewitt et al 24, Shapira et al 25, Park et al 24)
- forecasting macroeconomic variables (Bybee 23, Faria-e and Leibovici 24, Carriero et al 25)

Example

- Fix a lottery: $\bar{z}$ with probability p, $\underline{z}$ with probability $1 - p$
- What is an agent's **certainty equivalent** for this lottery?
  - i.e., the sure dollar amount that is considered equivalent to the random outcome of the lottery

The agent is an **expected utility maximizer** with utility function $v_\theta(z) = z^\theta$ where θ governs the degree of risk aversion.

In this model, the certainty equivalent of lottery $(\bar{z}, p; \underline{z}, 1 - p)$ is

$$v_\theta^{-1}\left(p v_\theta(\bar{z}) + (1 - p) v_\theta(\underline{z})\right)$$

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COMPUTER SCIENTIST

Use data of lotteries and certainty equivalents to train a machine learning algorithm (e.g., a neural net).

lottery
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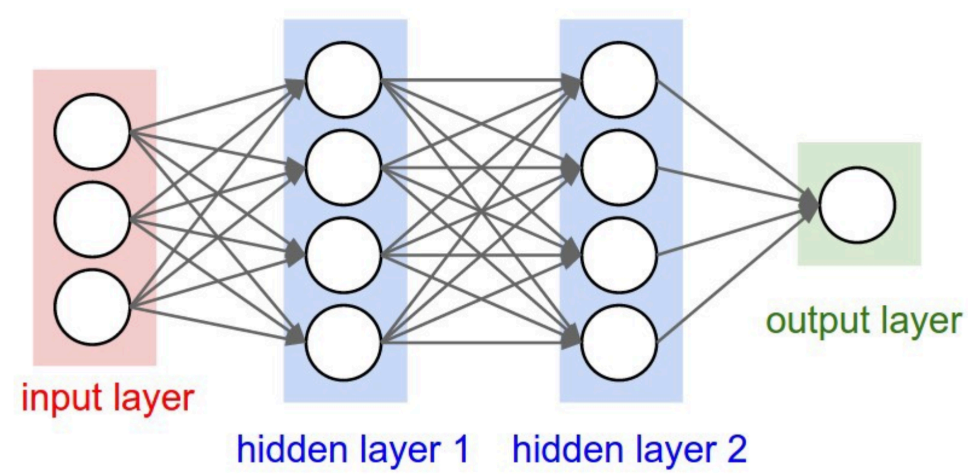

certainty equivalent y

ECONOMIST

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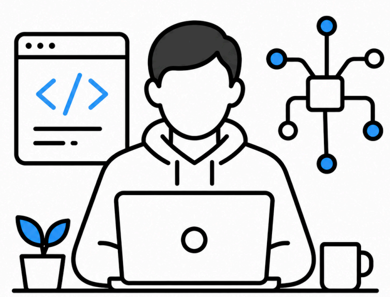

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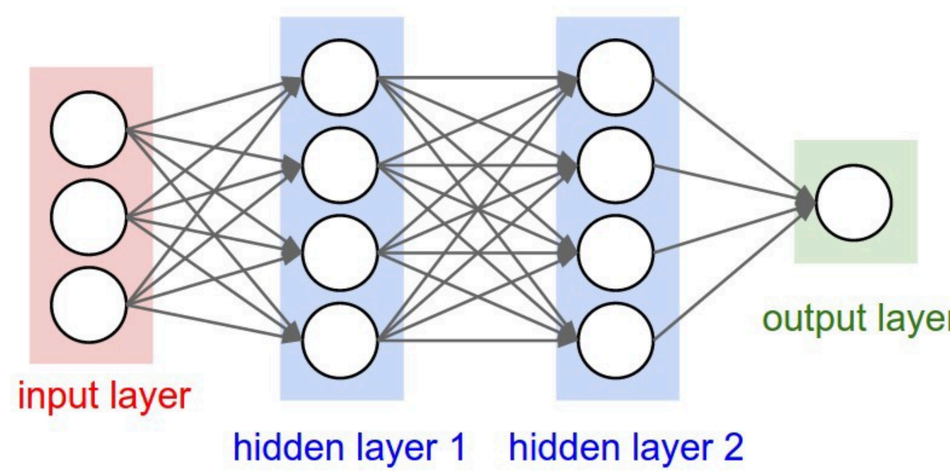

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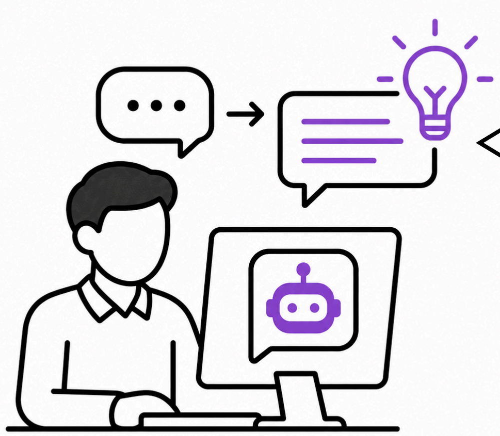
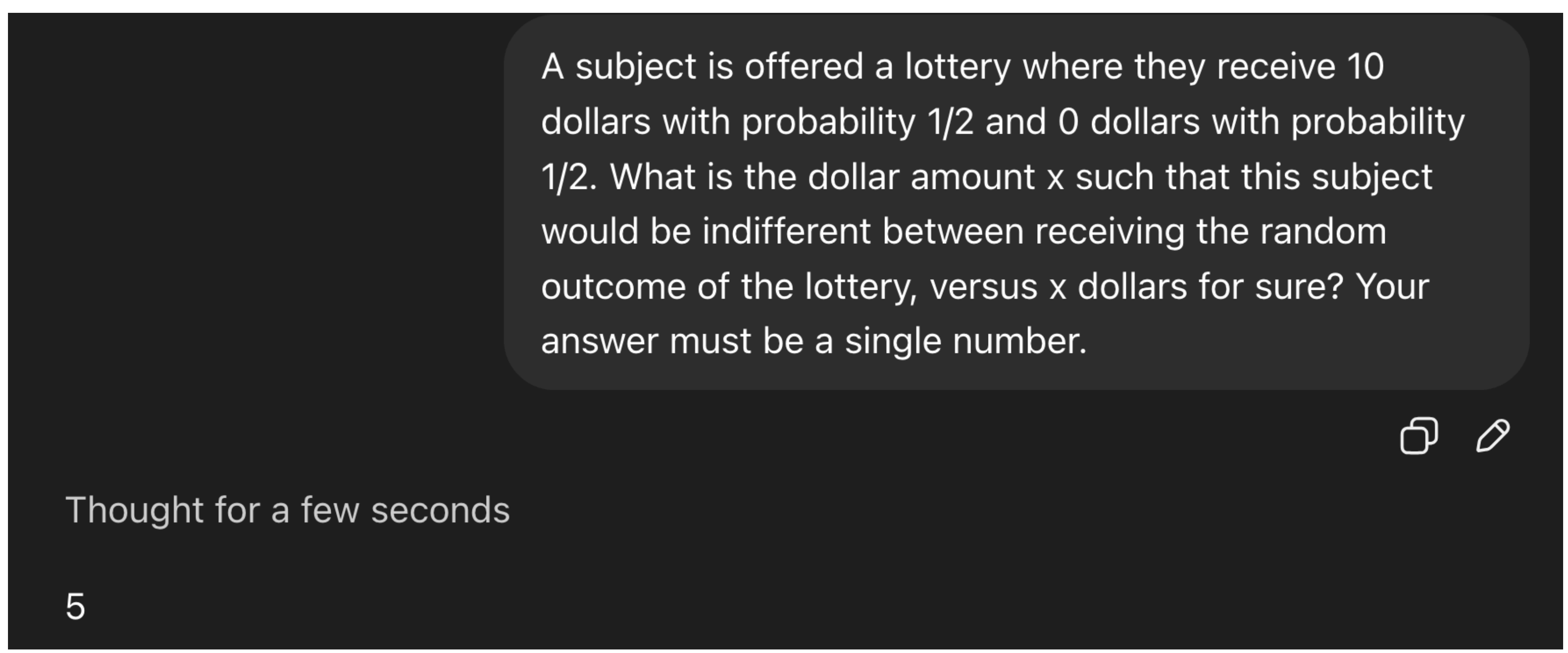


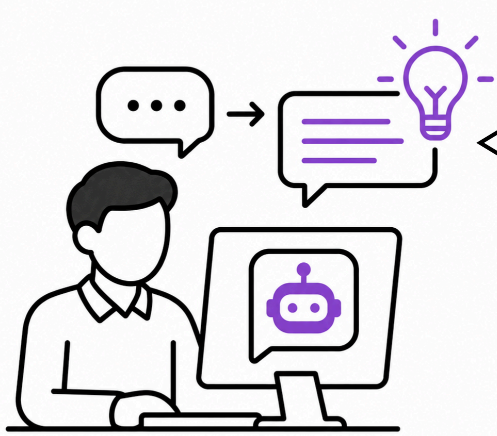
User of LLMs



Example

NO DOMAIN-SPECIFIC DATA REQUIRED!

...but are these predictions any good?



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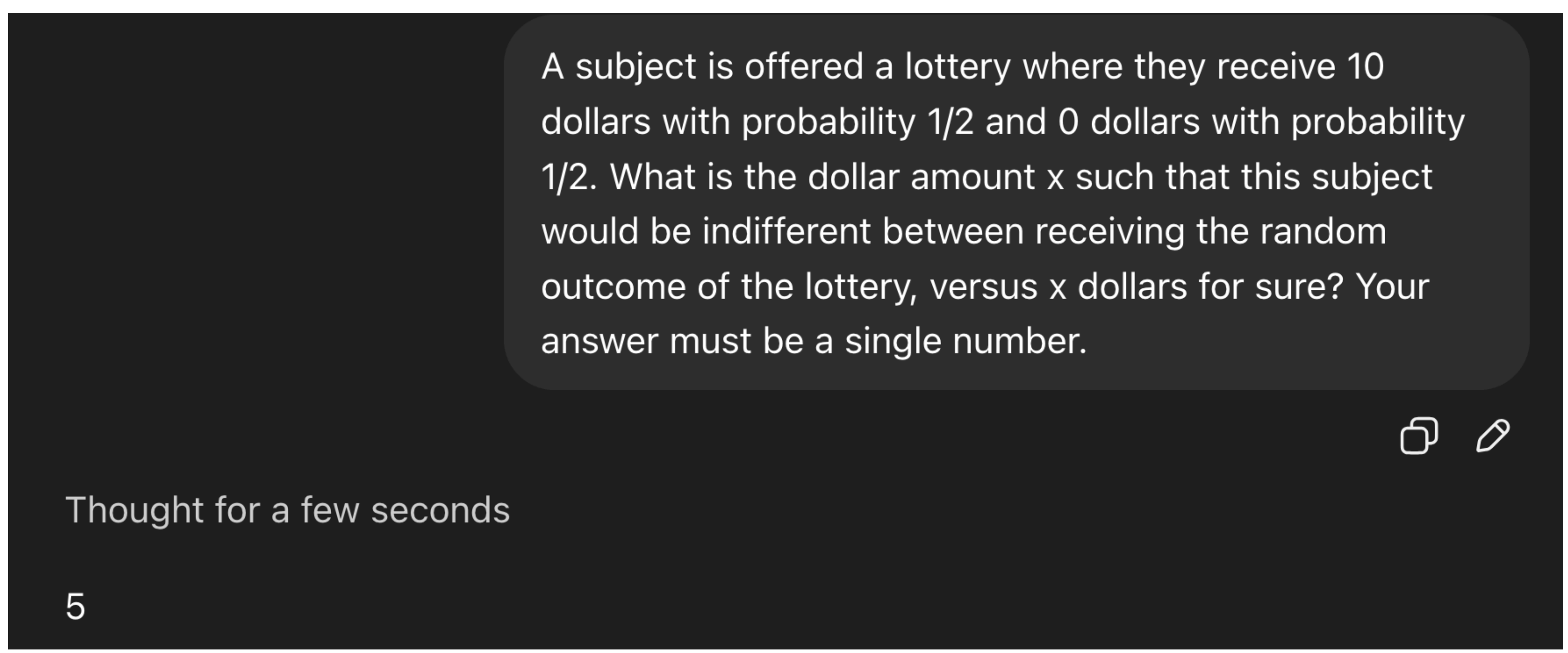

Proposed Measure

- We need a way to evaluate how useful the LLM is
- One option is simply "predictive accuracy"
- Two issues with this:
  - Predictive accuracy can be hard to interpret (is 2.4 RMSE small?)
  - The LLM exists in the context of alternative approaches to prediction, so its **relative** value is often what is relevant
- Key tradeoff: with enough domain-specific data, flexible algorithms will outperform the LLM, but that data is costly
- Our proposed measure: the LLM's **equivalent sample size**, defined as how much data is needed for the algorithm to beat the LLM
- Can be interpreted as a proxy for "cost saved"

This Talk

1. Framework and proposed measure

2. Statistical inference procedure
   - compare the LLM's prediction error to that of flexible ML algorithms/economic models trained on samples of increasing size

3. Applications
   - experimental behaviors: risk preferences, initial play in games
   - survey outcomes in PSID: wages, home-ownership, smoking, etc.

Framework

Prediction Problems

Generic **prediction problem**:

- covariates $X \in \mathscr{X}$
- outcome of interest $Y \in \mathscr{Y}$ (could be a distribution)

A **prediction rule** is a mapping $f : \mathscr{X} \to \mathscr{Y}$

Prediction Problems

Measure prediction accuracy via a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$

- e.g. squared error $\ell(y, \hat{y}) = (y - \hat{y})^2$ (regression)
- e.g., misclassification $\ell(y, \hat{y}) = 1\{\hat{y} \neq y\}$ (classification)

The **risk** (expected loss) of prediction rule f is

$$e(f) := \mathbb{E}_P[\ell(f(X), Y)]$$

where P is the distribution of (X, Y)

LLM Prediction Rule

Fix a pretained LLM and prompting protocol:

- Describe the prediction task
- Give the LLM X in natural language and ask it to predict Y

Denote the resulting prediction rule by f_{LLM} and its risk by $e_{LLM} \equiv e(f_{LLM})$

Domain-Specific Learning

An **algorithm** (or **economic model**) maps data to a prespecified set of prediction rules:

$$a : \mathscr{D} \to \mathscr{F}$$

where $\mathscr{D} \equiv \cup_{N \geq 1} (\mathscr{X} \times \mathscr{Y})^N$ is the set of all finite datasets.

- e.g., use the data to estimate the parameters of an economic model or ML algorithm by minimizing empirical risk

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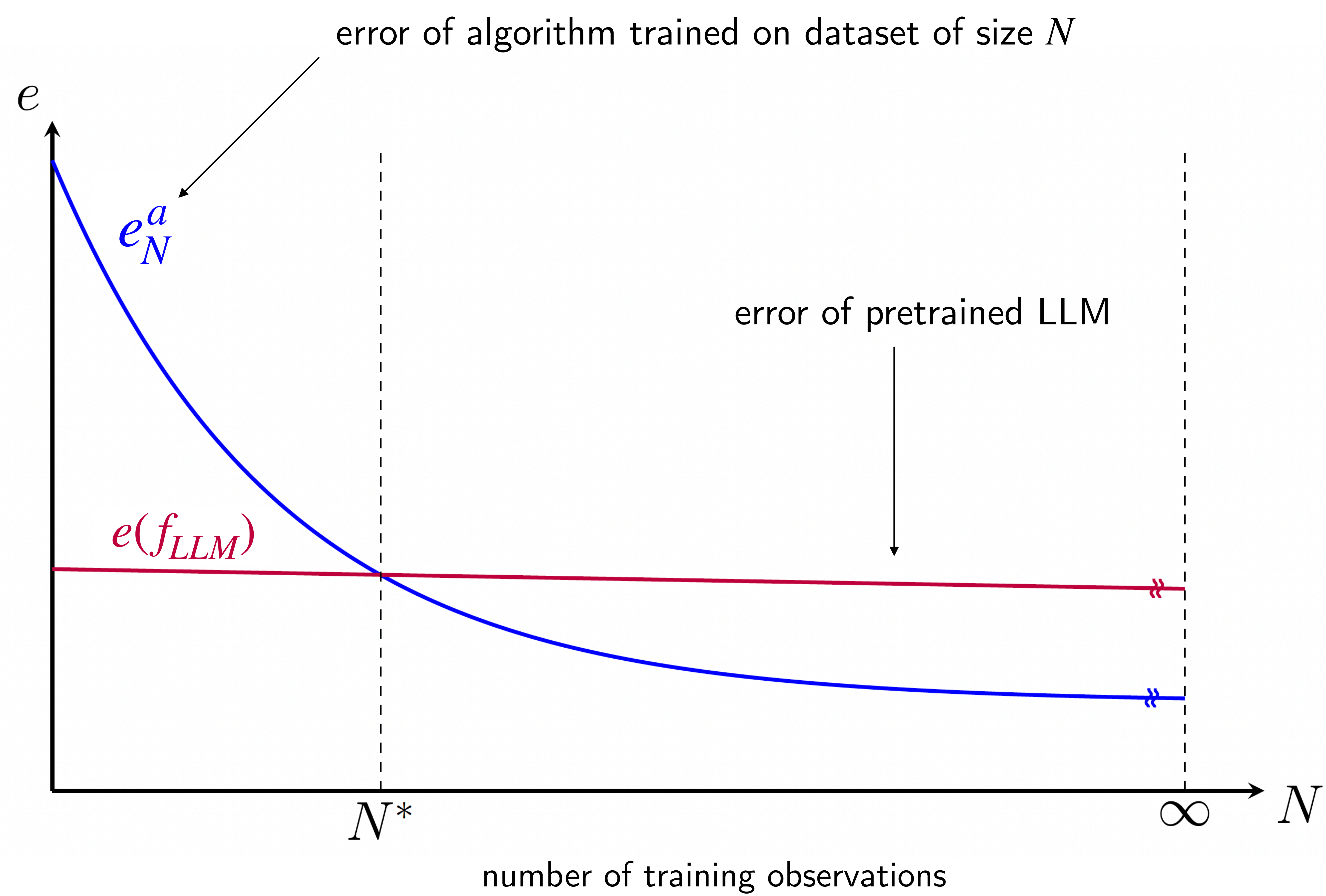
Given sample $D_N \equiv \{(x_i, y_i)\}_{i=1}^N$ define $f_N \equiv a(D_N)$ to be the (random) prediction rule we get from applying a to a (random) dataset of size N.

Define $e_N^a \equiv \mathbb{E}[e(f_N)] = \mathbb{E}_P[\ell(f_N(X, Y))]$ to be the expected risk of f_N.

Domain-Specific Error Curve



a-Equivalent Sample Size (ESS)

Definition: The LLM's a-**equivalent sample size (ESS)** is

$$N^* = \min\{N : e_N^a \leq e_{LLM}\}$$

- smallest training size such that the algorithm/model outperforms the pretrained LLM
- interpret as the minimum amount of domain-specific data the analyst would need to beat the LLM
- large N^* $\Rightarrow$ the LLM provides substantial predictive value in this domain; small N^* $\Rightarrow$ it does not

a-Equivalent Sample Size (ESS)

ESS depends on:

- The LLM and prompting protocol
- The problem domain
- The comparator algorithm a

Varying these and computing ESS for different choices can be instructive, as we will see in the applications

Estimation and Inference for N^*

Inference on N^*

- Formulate a sequence of hypotheses tests:

$$H_{0,N} : e_N^a \leq e_{LLM} \quad N = 1, 2, \ldots$$

  i.e., the algorithm trained on N observations beats the LLM

- We provide a valid statistical test for each null $H_{0,n}$

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- We provide a valid statistical test for each null $H_{0,n}$
- Then propose a sequential procedure:
  - Test $H_{0,1}$ at confidence level α, if reject then move onto $H_{0,2}$, etc...
  - At first N at which we cannot reject, stop and set $\widehat{N}_\alpha^* = N$
- Finally argue that validity of the individual tests implies that $\left[\widehat{N}_\alpha^*, \infty\right)$ is a valid one-sided confidence interval for N^* at level α

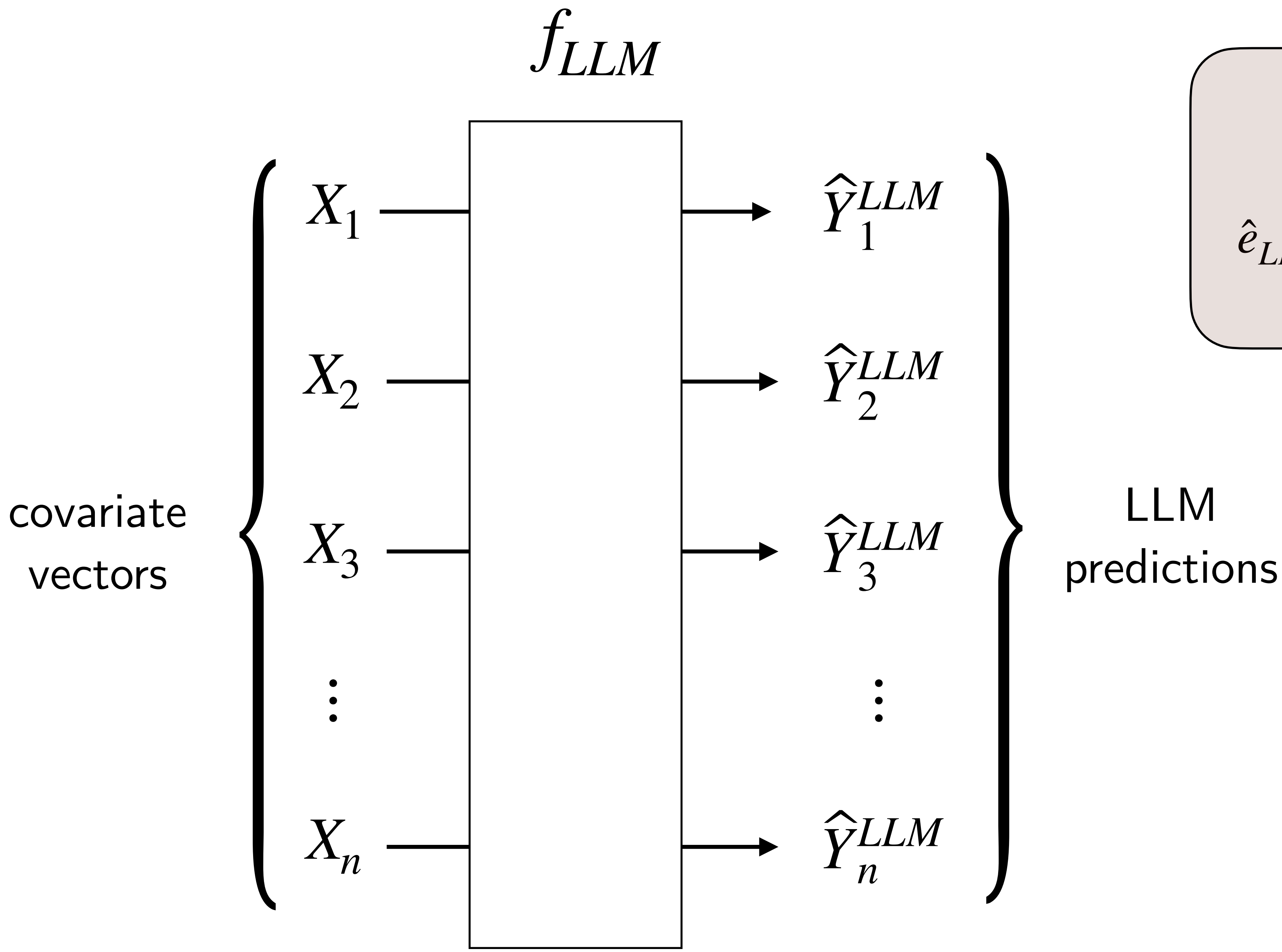
Test for Each $H_{0,k}$

- Estimating e_{LLM}: compute empirical counterpart $\widehat{e}_{LLM}$
- Estimating $e_N := e_N^a$: develop a block-out cross-validation estimator $\widehat{e}_N^{CV}$
- Define the test statistic $T_{N,n} \equiv \dfrac{\widehat{e}_N^{CV} - \widehat{e}_{LLM}}{\sqrt{\widehat{\sigma}_N / n}}$ where $\widehat{\sigma}_N^2$ is an estimator of the variance of $\widehat{e}_N^{CV} - \widehat{e}_{LLM}$ (see paper)

Reject $H_{0,N}$ if $T_{N,n} > c_\alpha$ for critical value c_α

Estimate of e_{LLM}

Given a **full sample** $(X_i, Y_i)_{i=1}^n$, feed each X_i to the LLM and obtain prediction $\widehat{Y}_i^{LLM}$

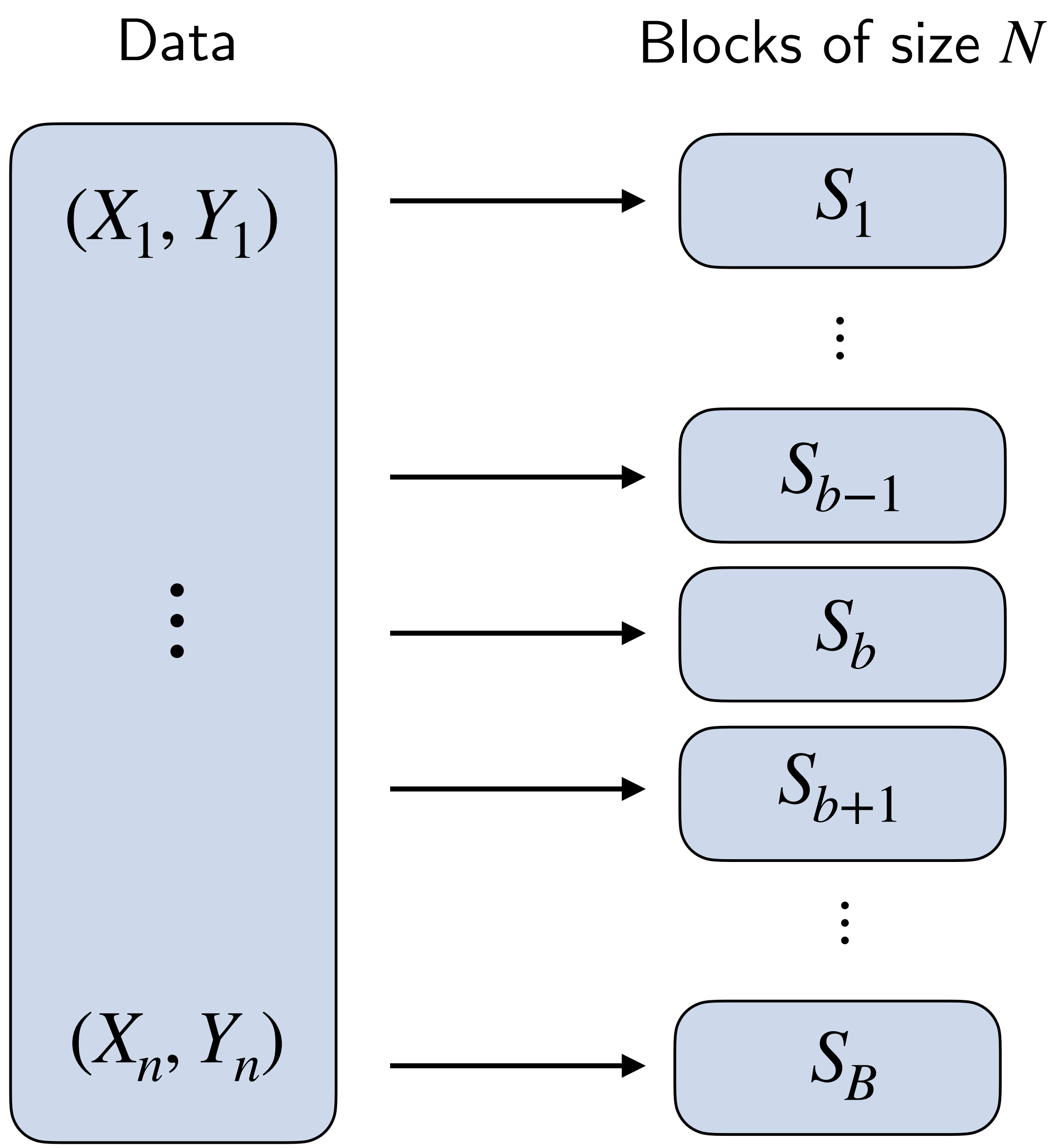


Empirical risk:

$$\hat{e}_{LLM} = \frac{1}{n}\sum_{i=1}^{n} \ell\left(\widehat{Y}_i^{LLM}, Y_i\right)$$

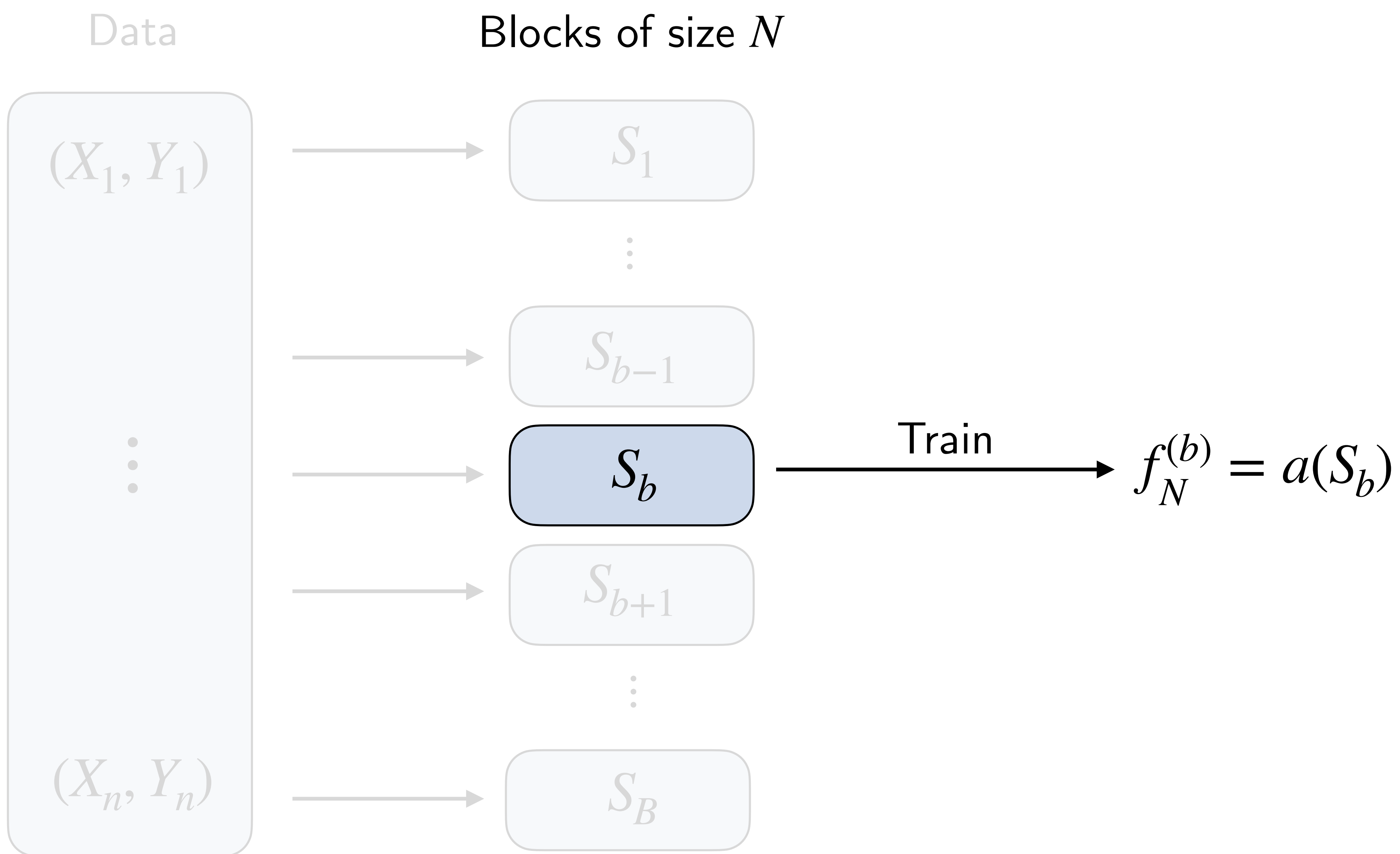
Estimate of e_N

1. Partition the **data** $(X_i, Y_i)_{i=1}^n$ into B disjoint blocks, **each of size** N



Estimate of e_N

2. For each block b, **train** the algorithm a on S_b and **test** on complement $S_b^c := \bigcup_{b' \neq b} S_{b'}$



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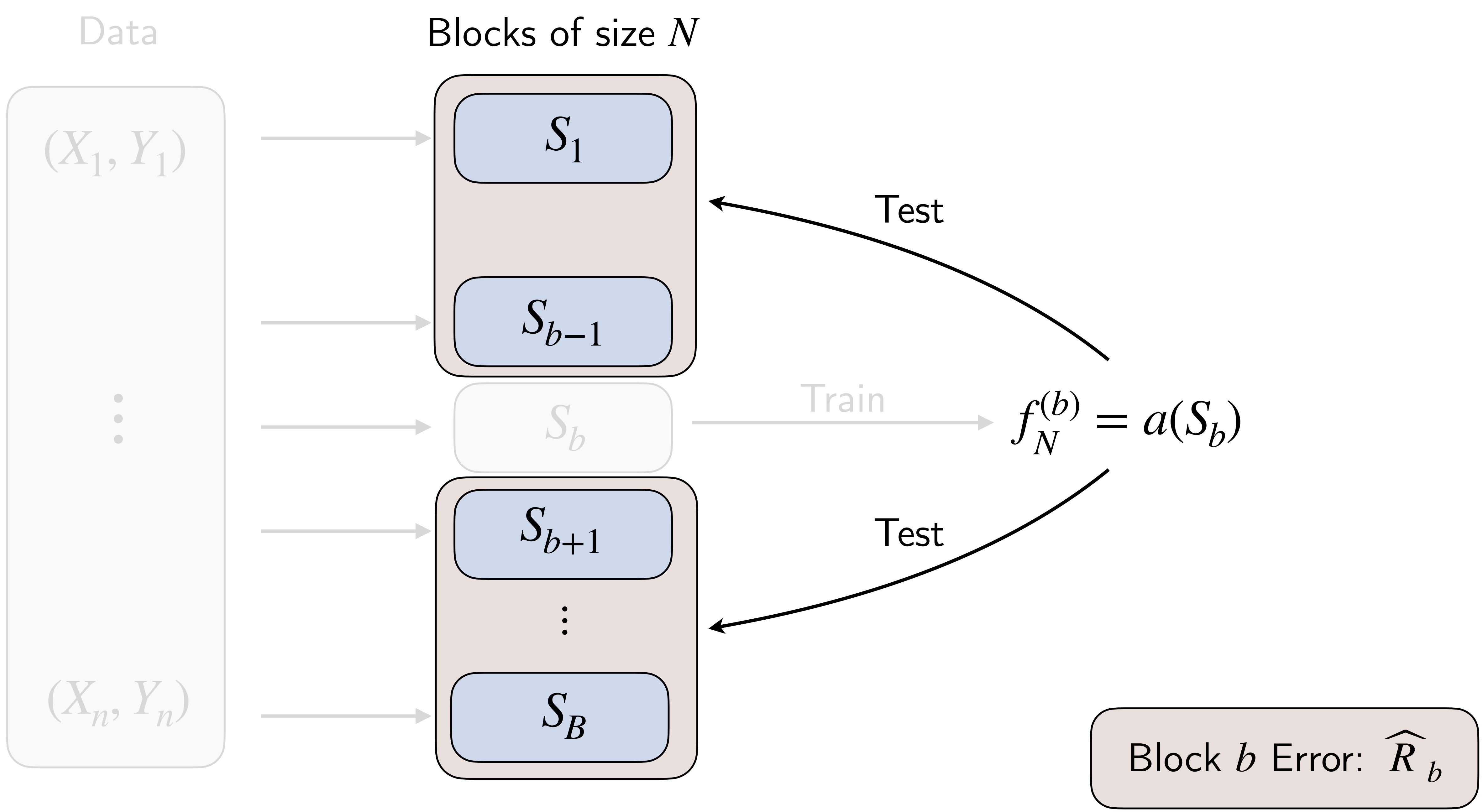
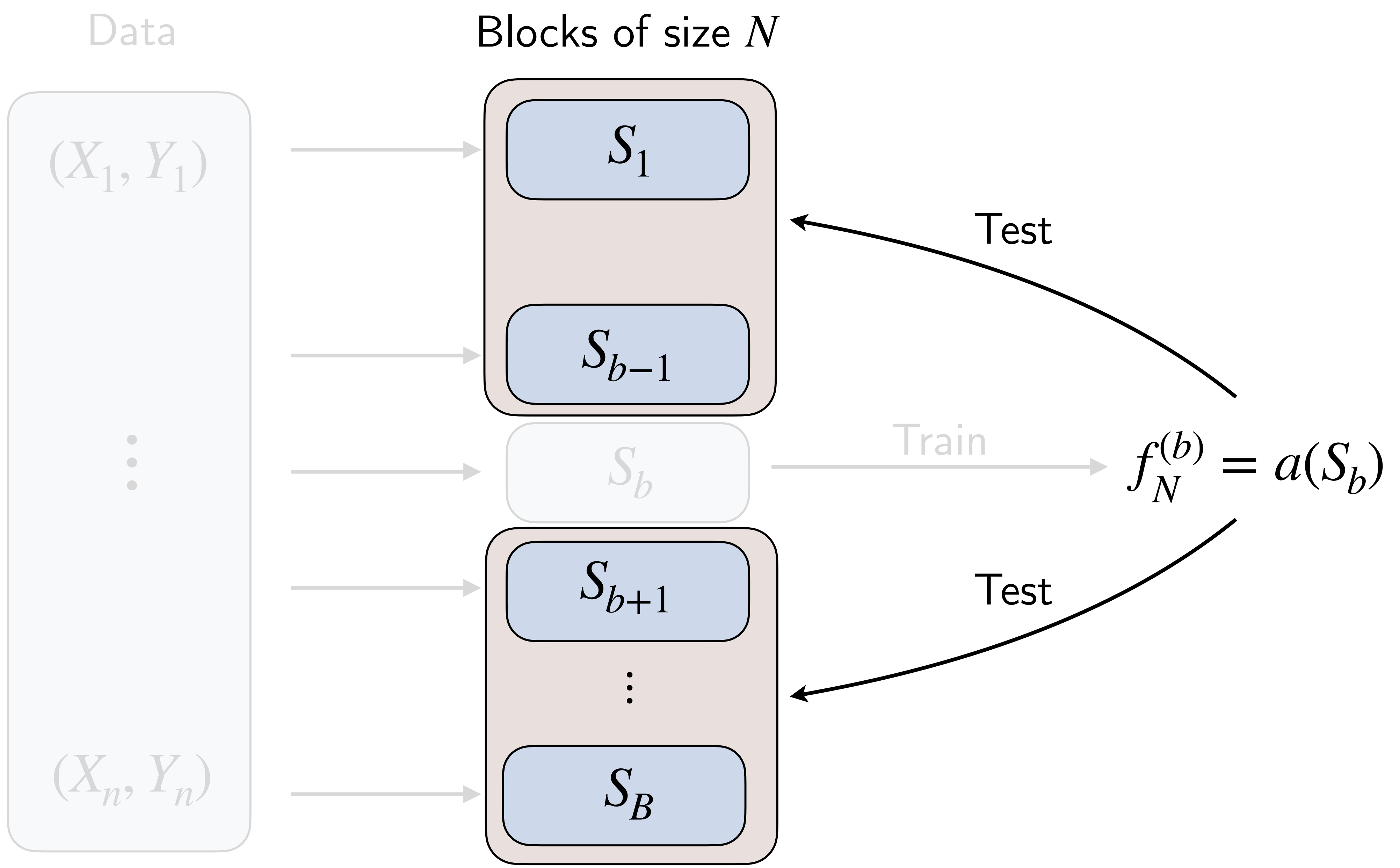


Estimate of e_N

3. Average the error over choice of training block

Estimate: $\hat{e}_N^{CV} := \frac{1}{B} \sum_{b=1}^{B} \widehat{R}_b$



Plugin Estimator for N^*

Plugin estimator for N^* given these estimates:

$$\widehat{N}^* = \min\left\{N : \hat{e}_N^{CV} \leq \hat{e}_{LLM}\right\}$$

i.e., smallest integer N such that our estimate of the algorithm's error given N datapoints is smaller than that of the LLM

Inference on N^*

- Recall our a sequential test:
  - Test $H_{0,1} : e_1^a \leq e_{LLM}$, if reject then move onto $H_{0,2}$, etc...
  - At first point at which we cannot reject, stop and set $\widehat{N}_\alpha^* = N$
- Proposed confidence interval: $[\widehat{N}_\alpha^*, \infty)$

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- Proposed confidence interval: $[\widehat{N}_\alpha^*, \infty)$
- By definition of N^*:

$$\underbrace{e_{N_1} > e_{LLM} \quad \cdots \quad e_{N^*-1} > e_{LLM}}_{\text{Null is FALSE}} \quad \underbrace{e_{N^*} \leq e_{LLM} \quad e_{N^*+1} \leq e_{LLM} \quad \cdots}_{\text{Null is TRUE}}$$

$$\text{FWER} = \Pr(\text{reject any true } H_{0,k}) = \Pr(\text{reject } H_{0,N^*}) \leq \alpha$$

Statement of Result

Theorem: Under standard regularity conditions (see paper), this procedure implemented with rejection rule $T_{N,n} > c_\alpha$ for each step N satisfies

$$\lim_{n\to\infty} FWER_n \leq \lim_{n\to\infty} \Pr\left(T_{\widehat{N}^*_\alpha,n} > c_\alpha\right) \leq \alpha$$

Application 1:

Predicting Risk Preferences

Setting

Prediction problem: Given a country and a lottery, predict the average certainty equivalent reported by subjects in that country

- e.g., "what are people willing to pay (on average) for the outcome of a lottery that is equally likely to yield $100 or $0?"

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Our question: How useful are LLMs for predicting certainty equivalents, and how does this vary across countries?

- We treat each of the 30 countries as a separate dataset
- Evaluate the LLM's ESS for each country's prediction problem

Cross-Country Data and LLM Predictions

LLM prediction: use GPT-5.4 to predict an average certainty equivalent for each lottery

```
In a cross-country experiment,
participants in {country} stated the guaranteed amount that
would make them indifferent between receiving that amount for
sure and playing a lottery.

For the lottery below, simulate certainty equivalents for
200 synthetic subjects drawn from the participant pool in
{country}. Treat the draws as different plausible respondents,
not repeated copies of the same person. Allow for realistic
heterogeneity across subjects in how they respond, but do not
force the draws to be evenly spaced or artificially diverse.
At the same time, do not collapse the distribution to nearly
identical values.

Lottery:
- Outcome 1: {z_1} with probability {p_1}
- Outcome 2: {z_2} with probability {1-p_1}

Return exactly 200 numeric certainty equivalents. Each draw
should be a plausible individual response for one subject.
```

LLM prediction: the average of the simulated certainty equivalents

Cross-Country Data and LLM Predictions

LLM prediction: use GPT-5.4 to predict an average certainty equivalent for each lottery

Domain-specific comparators: we consider a random forest algorithm and two popular economic models

- Expected Utility with utility function $u(z) = z^{\alpha}$ for gains, $u(z) = -(-z)^{\beta}$ for losses
- Cumulative Prospect Theory: agents maximize expected utility but distort probabilities

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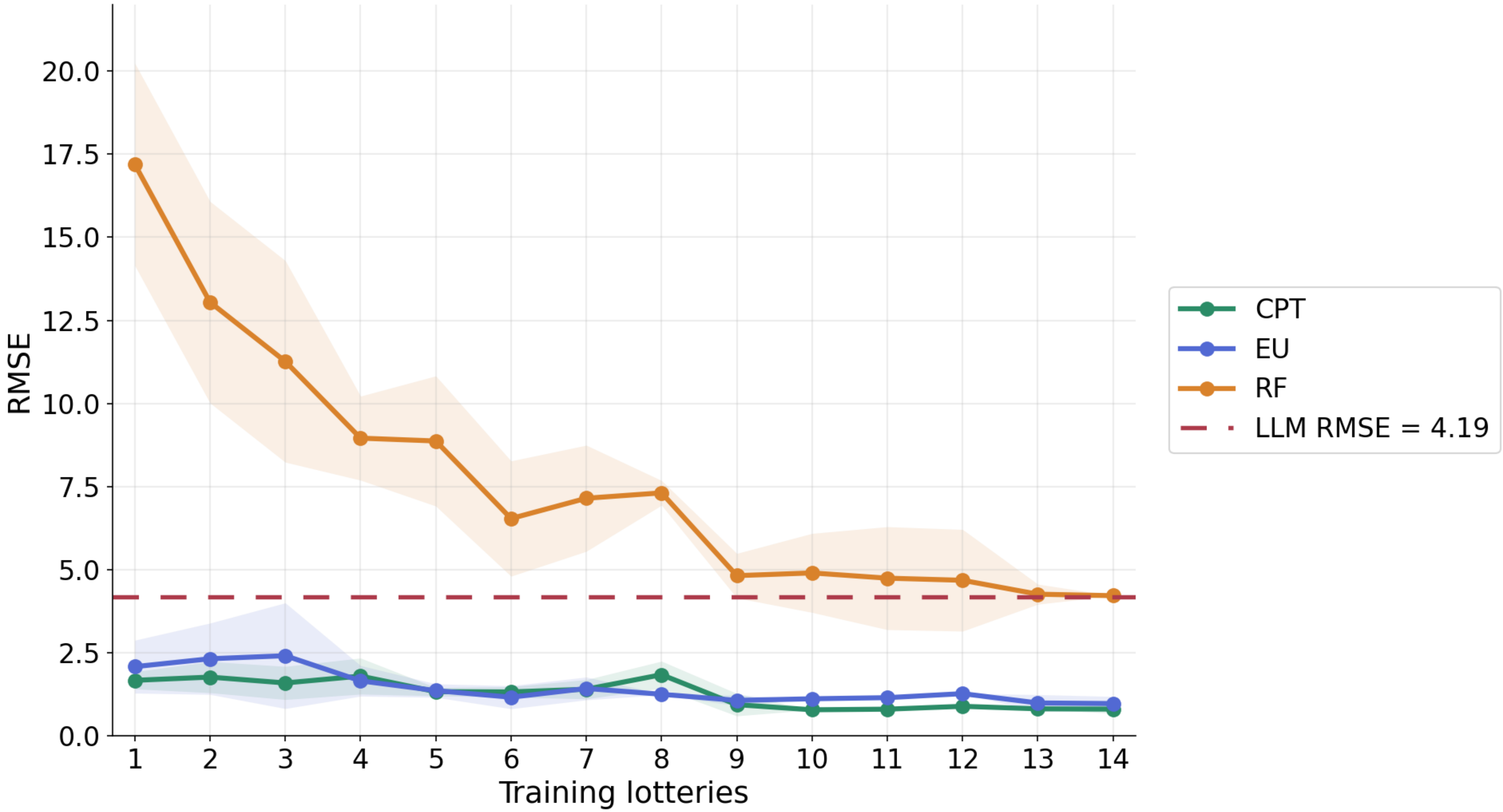
ESS tells us: how many training lotteries do the domain-specific comparators need to beat the LLM's performance?

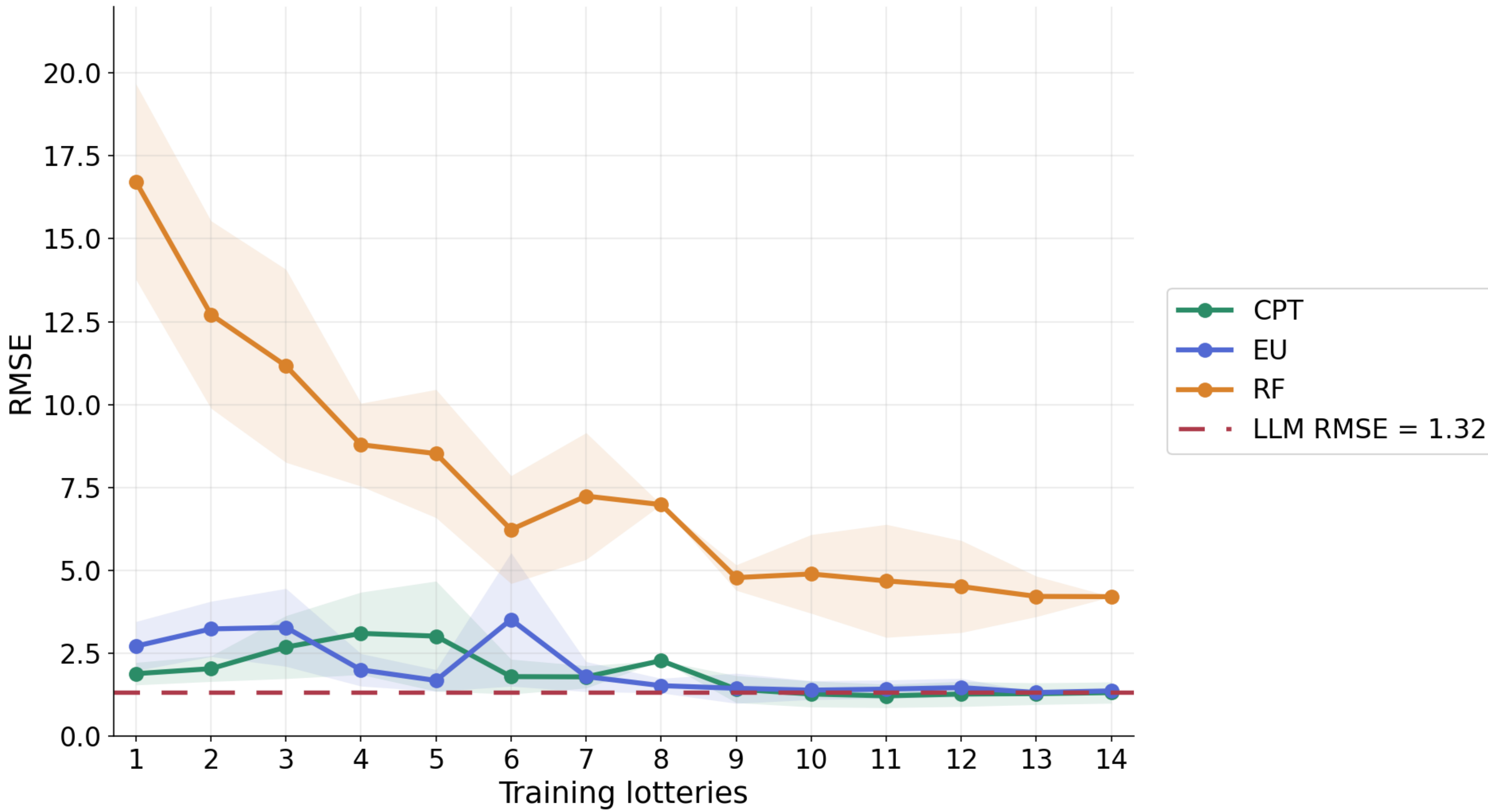
(Weakly) Lowest ESS Country: Japan



The Econ models beat GPT 5.4 when trained on a single lottery!

RF takes 14 lotteries to catch up

Highest ESS Country: Germany

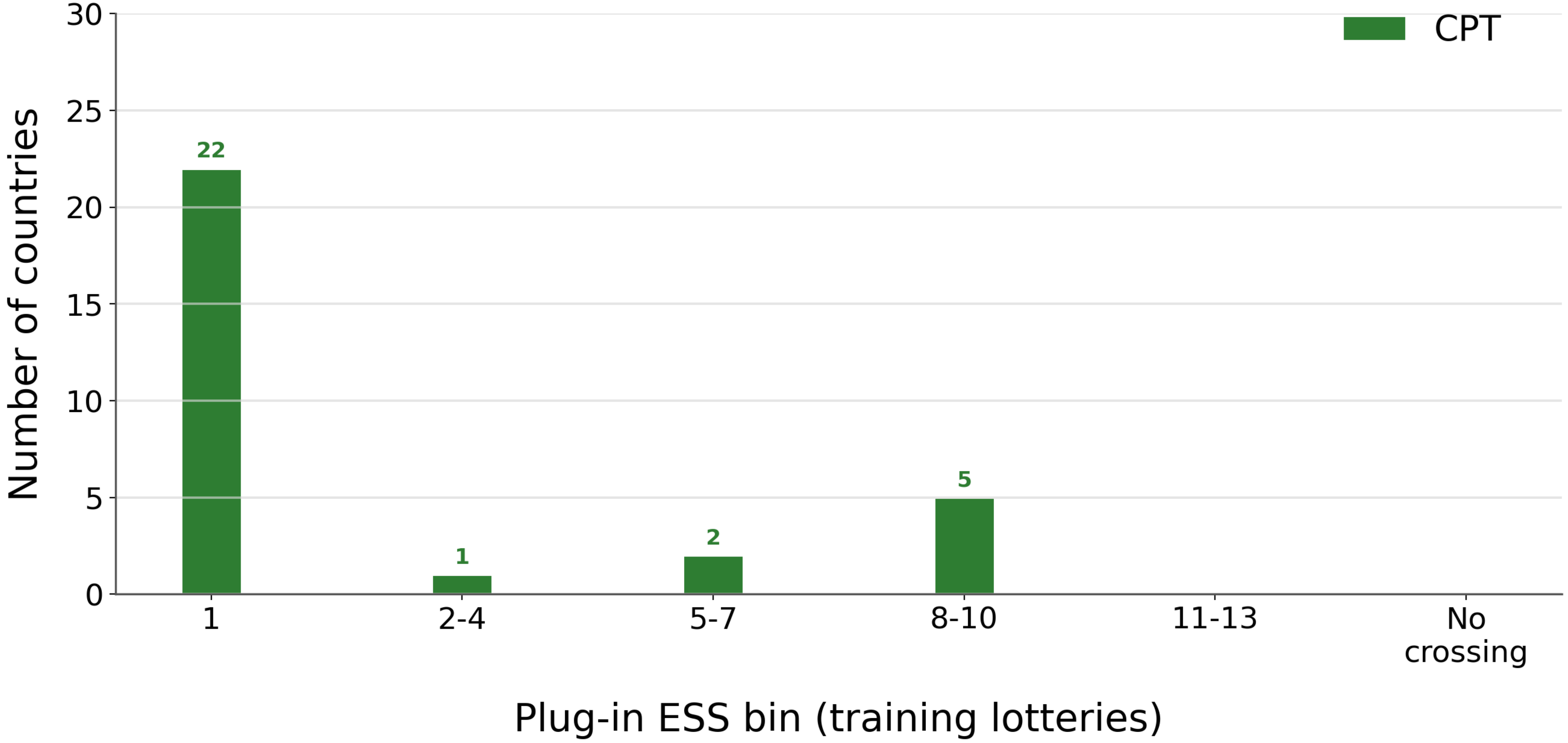


The Econ models need more lotteries here to beat GPT 5.4

RF never catches up in the considered domain

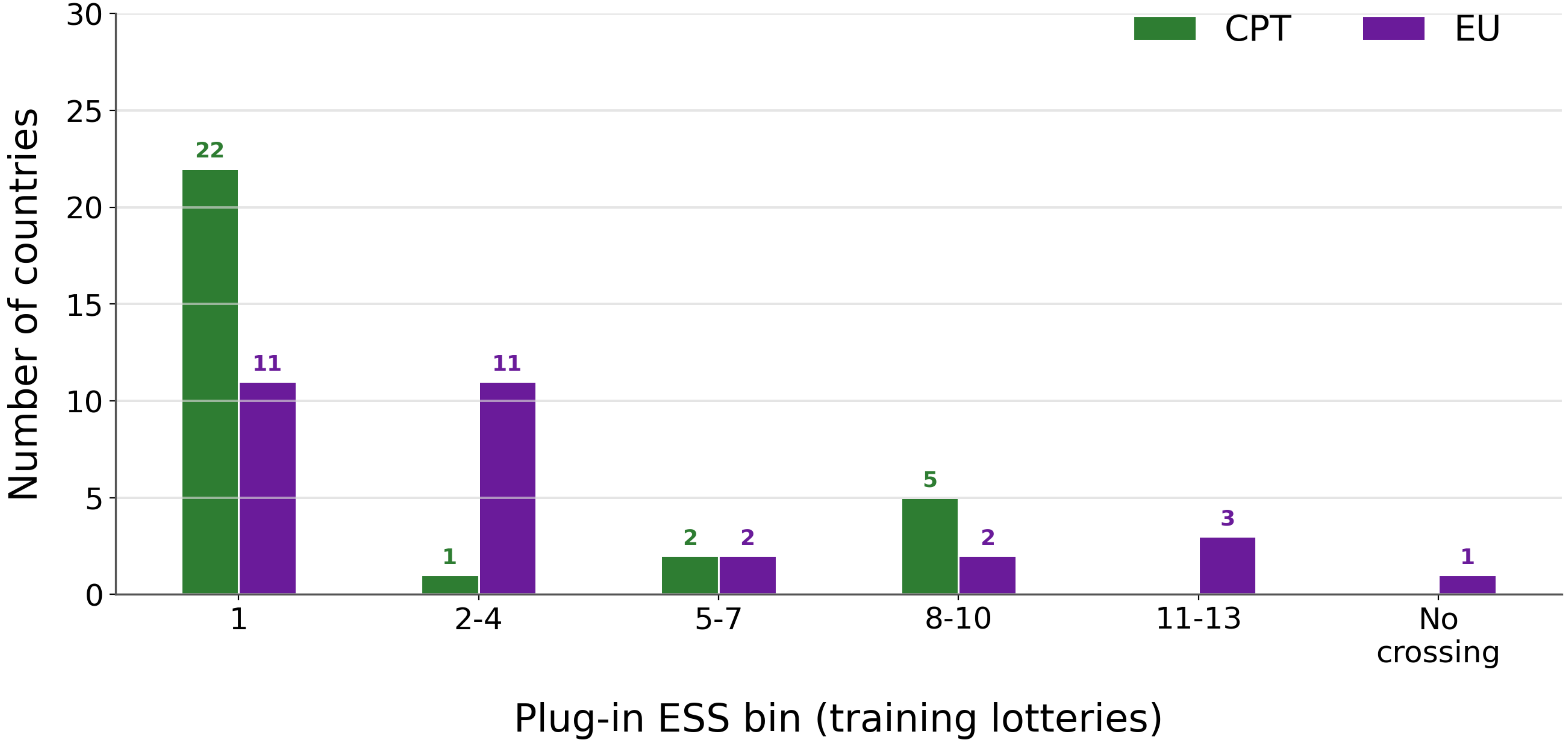
ESS Across Countries

Median CPT ESS = 1



ESS Across Countries

Median CPT ESS = 1 Median EU ESS = 4



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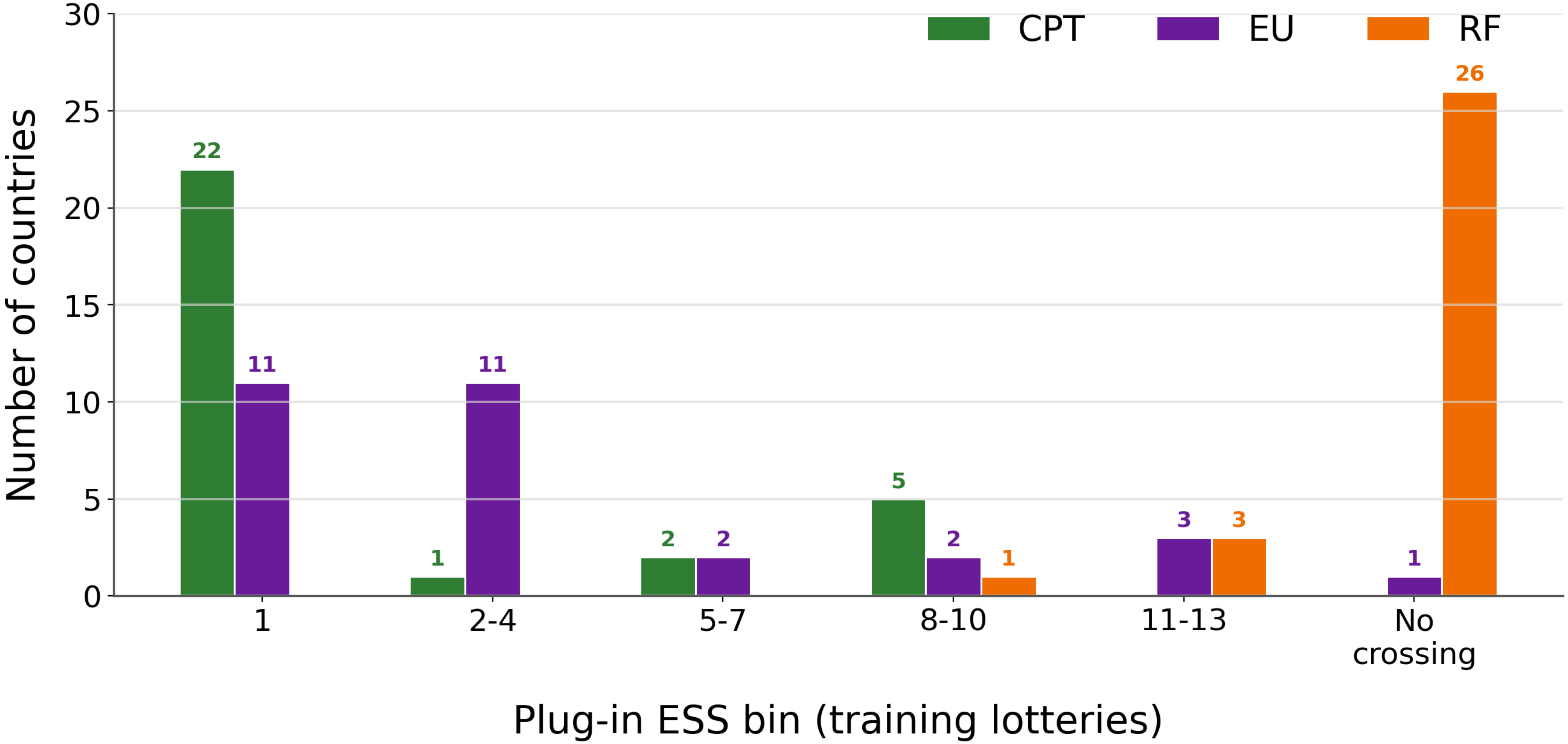

Takeaways: ESS is generally low for the economic models, while RF often does not cross, but there is variation across countries

What Drives the Variation?

- The UN Human Development Index (HDI) is a composite of life expectancy, education, and per-capita income, normalized between 0 and 1
- LLM error tends to be lower in higher-HDI countries ($\rho = -0.44$), with a substantially weaker relationship for CPT ($\rho = -0.19$)
- ESS tends to be higher in higher-HDI countries ($\rho = 0.34$)
- Takeaway: LLMs are relatively more useful for prediction in more developed countries
  - Intuitive, but our measure helps us see this more clearly

Application 2:

Predicting Play in Games

Setting

Classification problem: given a 3x3 payoff matrix, predict the action most likely to be chosen by the row player.

	a_1	a_2	a_3
a_1	25, 25	30, 40	100, 31
a_2	40, 30	45, 45	65, 0
a_3	31, 100	0, 65	40, 40

X

$$\longrightarrow \quad \underset{Y}{\{a_1, a_2, a_3\}}$$

Our analysis: evaluate the ESS of GPT 5.4 relative to:

- A popular behavioral game theory model (Poisson Cognitive Hierarchy)
- Random Forest

Data

We consider two datasets:

- **Dataset A**: play in 86 games, aggregated across six experimental game theory papers (Wright and Leyton-Brown 13)

Data

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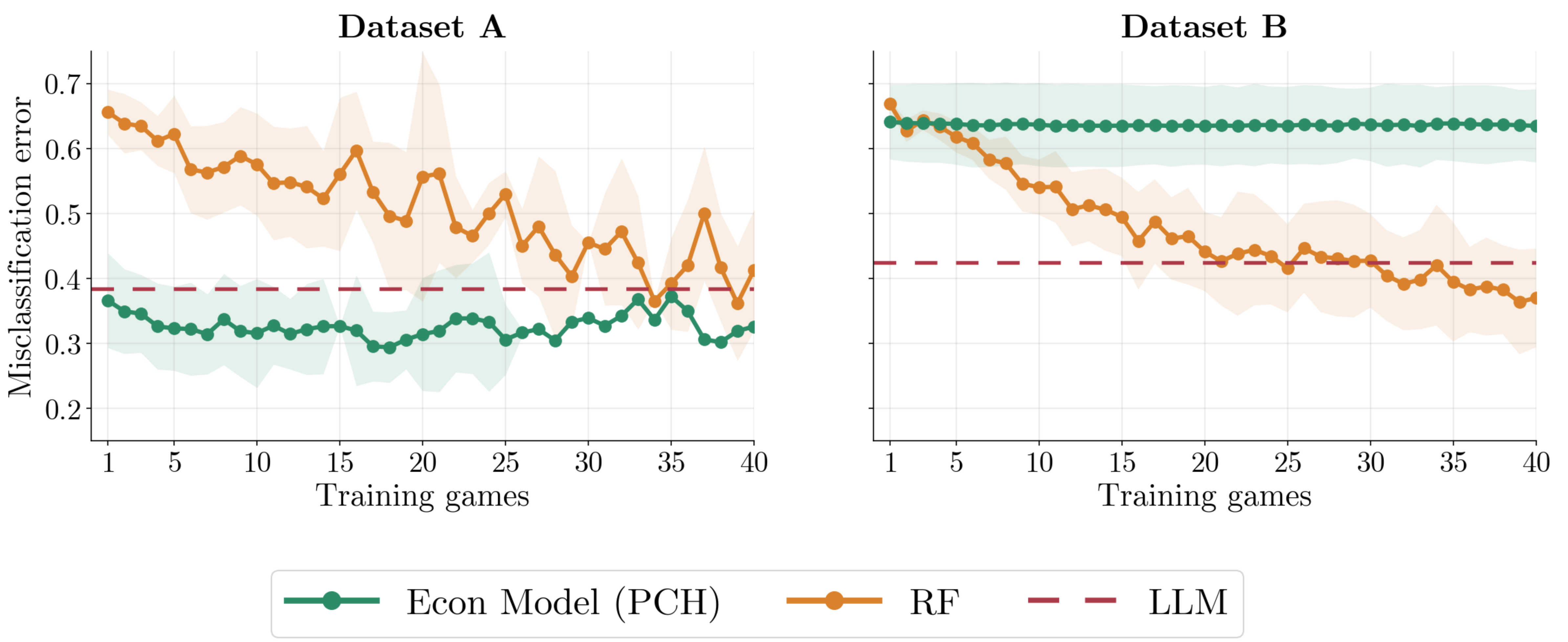
- **Dataset A**: play in 86 games, aggregated across six experimental game theory papers (Wright and Leyton-Brown 13)
- **Dataset B:** play of 200 games, algorithmically generated to be strategically rich and different from those in Dataset A (Fudenberg and Liang 19)

Example:

	a_1	a_2	a_3
a_1	$90, 90$	$30, 80$	$45, 30$
a_2	$80, 30$	$55, 55$	$37, 5$
a_3	$30, 45$	$5, 37$	$70, 70$

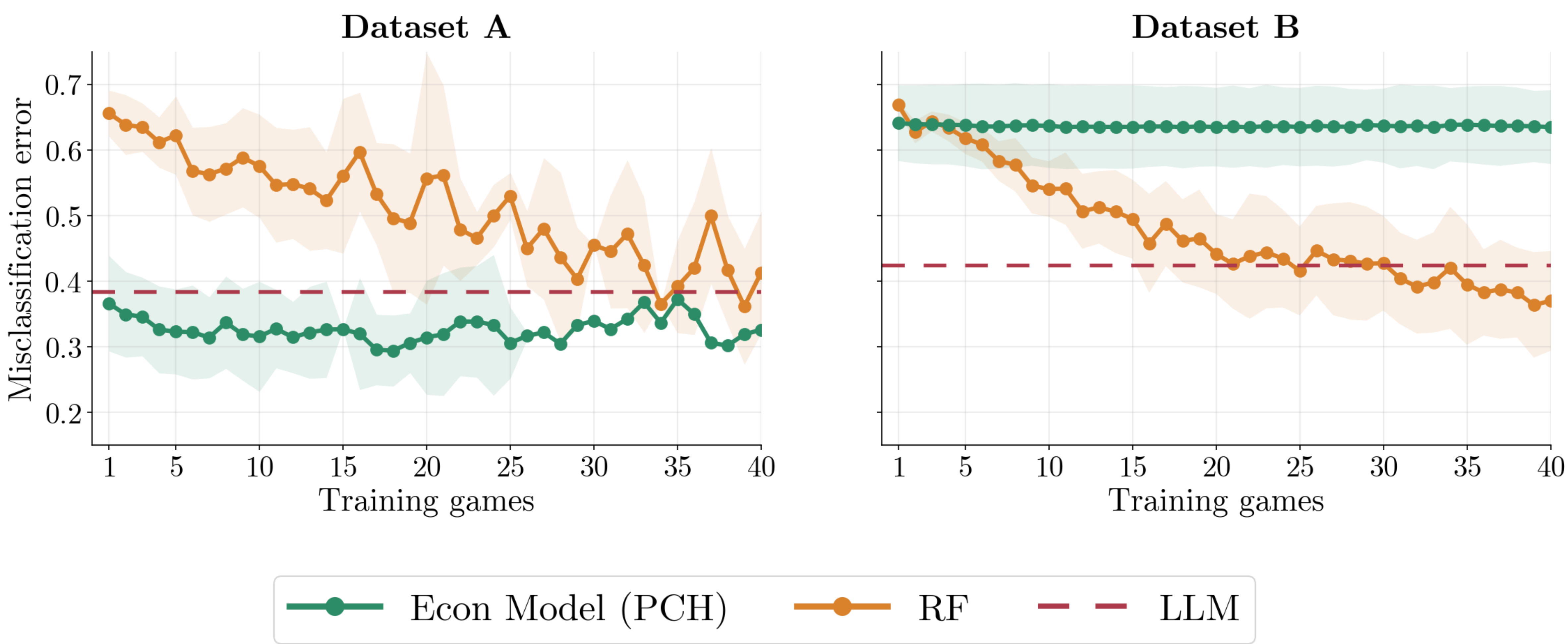
Action a_2 maximizes expected payoffs against uniform play, but but (a_1, a_1) is a Pareto-dominant Nash equilibrium

Data



Dataset A
Dataset B
Misclassification error
0.7
0.6
0.5
0.4
0.3
0.2
1
5
10
15
20
25
30
35
40
Training games
Econ Model (PCH)
RF
LLM

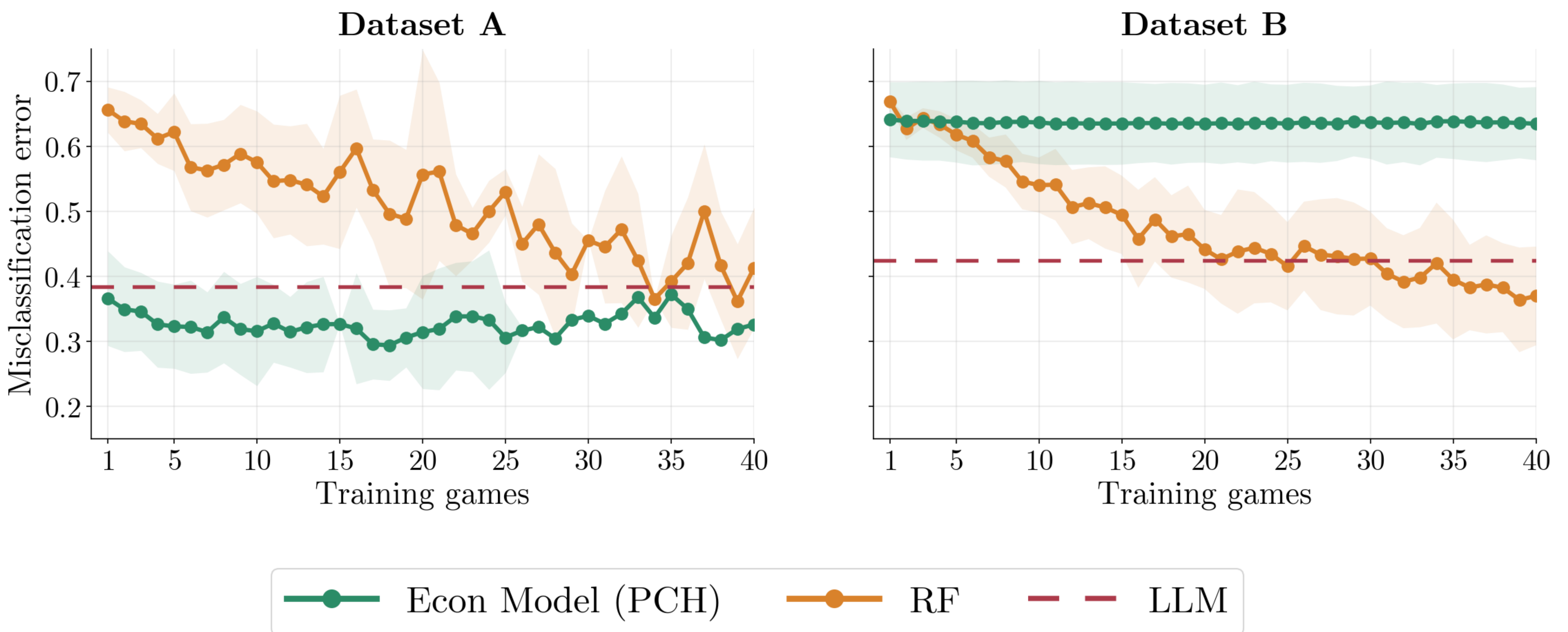
Data



On the lab games designed by experimental game theorists:

- ESS very low with respect to the economic model (Poisson Cognitive Hierarchy Model): plugin estimate $\widehat{N}* = 1$, confidence interval of $[1,\infty)$

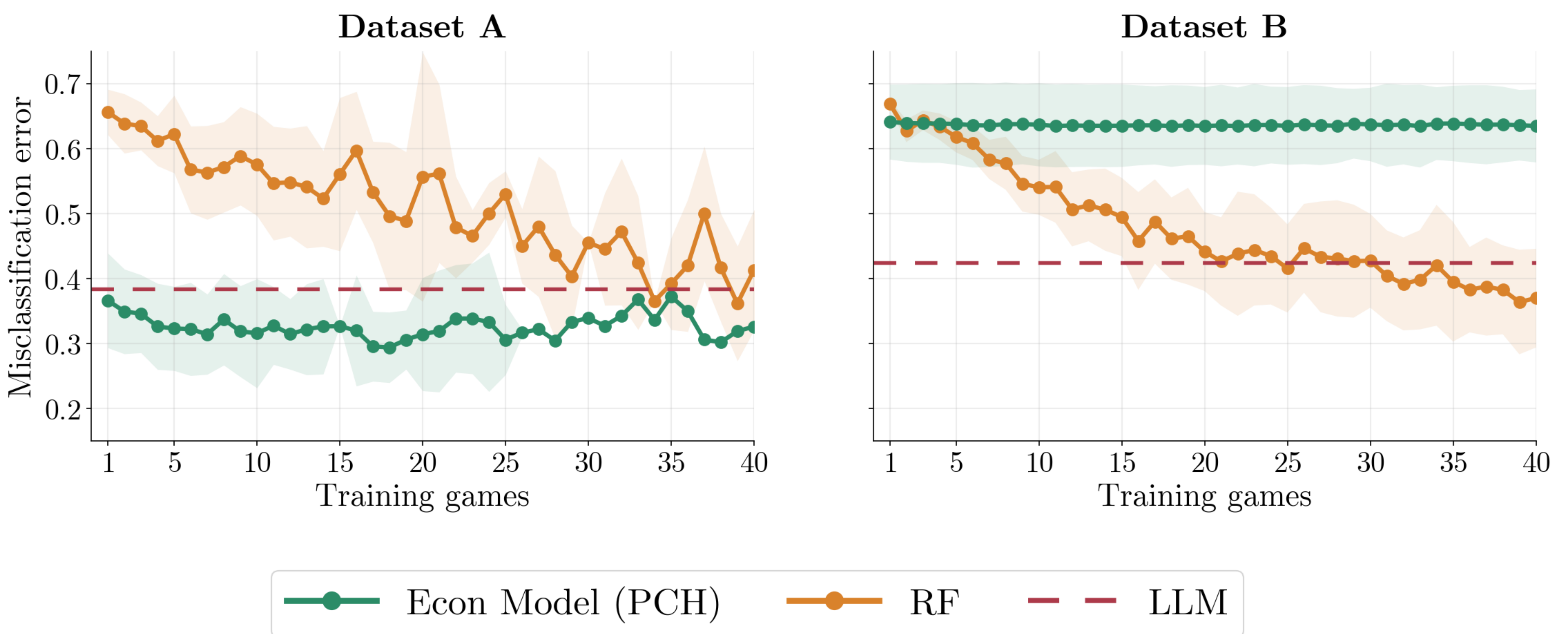
Data



In the new games:

- The LLM does nearly as well as it did on the lab games, but the economic model does much worse, and stays worse
- Eventually the ML algorithm surpasses the LLM (plugin estimate $\widehat{N}* = 25$, confidence interval of $[16,\infty)$)

Data



Takeaway: LLM prediction is more useful when our theories fail

Application 3:

Survey Outcomes in PSID

Application 3: PSID

We predict outcomes in the Panel Survey of Income Dynamics (PSID)

- Longitudinal survey that has followed US individuals and families since 1968, with data on economic, social, and health factors

```
Here is your persona:  You are a 37-year-old
male living in Tennessee in 2020.  You work in
architecture and engineering occupations in
manufacturing industry.  You have 4 years of
work experience in the current job.  You have
completed 14 years of education.  You are
married and have 1 child.  Your father has 12
years of education.  Your mother has 12 years
of education.  Your race is white.  Your health
is excellent.  You are Protestant.

Given your persona, do you own a house or
apartment or do you rent?  If you own, say 1;
if you rent, say 5; if you neither own nor
rent, say 8.
```

Application 3: PSID

We predict outcomes in the Panel Survey of Income Dynamics (PSID)

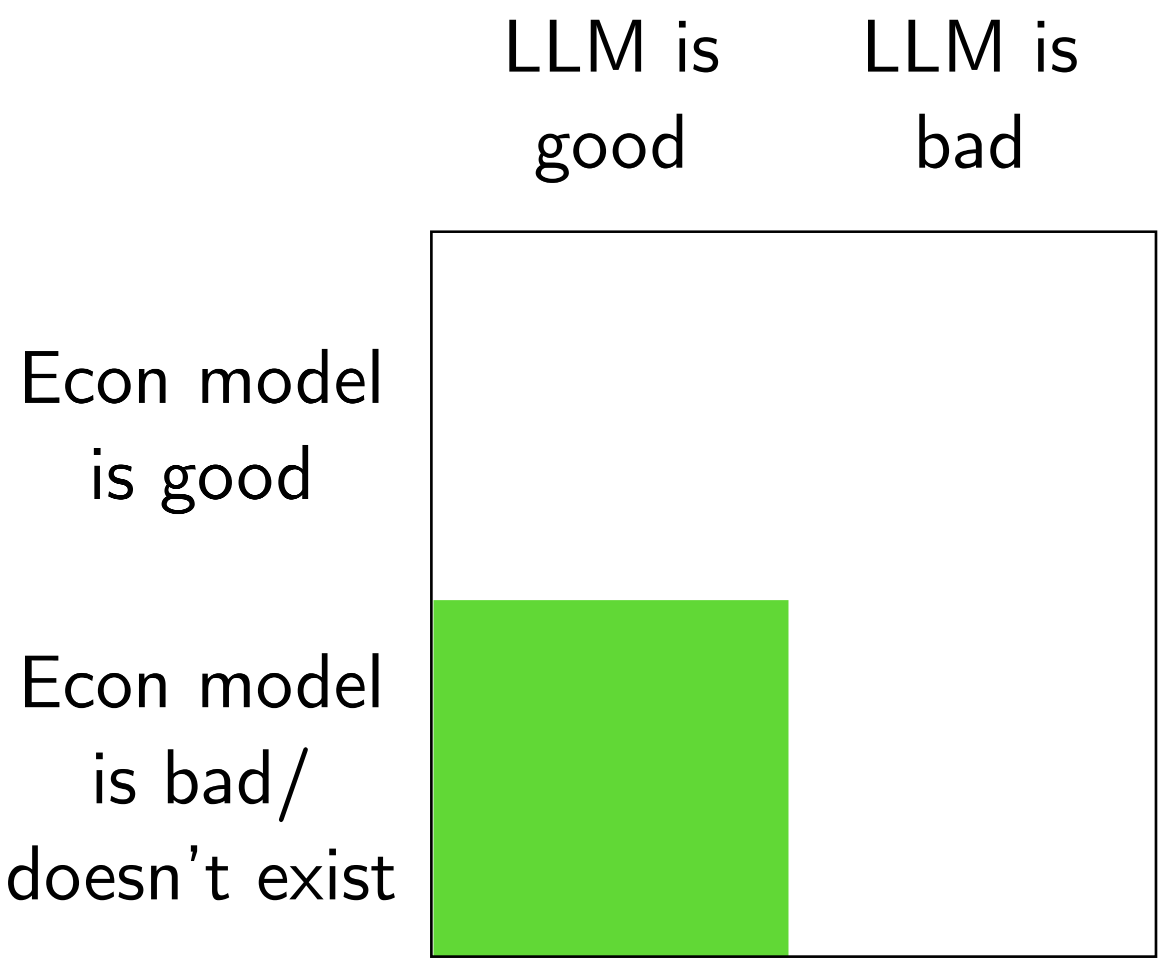
- Longitudinal survey that has followed US individuals and families since 1968, with data on economic, social, and health factors
- Consider a PSID year beyond the LLM's knowledge cutoff
- Main takeaway: ESS (computed using RF) differs substantially across different outcome variables, e.g.,
  - **Smoking**: one-sided CI is $[1,\infty)$, plugin estimator $\widehat{N}^* = 1$
  - **Hourly wage:** one-sided CI is $[17,\infty)$, plugin estimator $\widehat{N}^* = 49$
  - **Home-ownership:** one-sided CI is $[590,\infty)$, no finite crossing

Conclusion

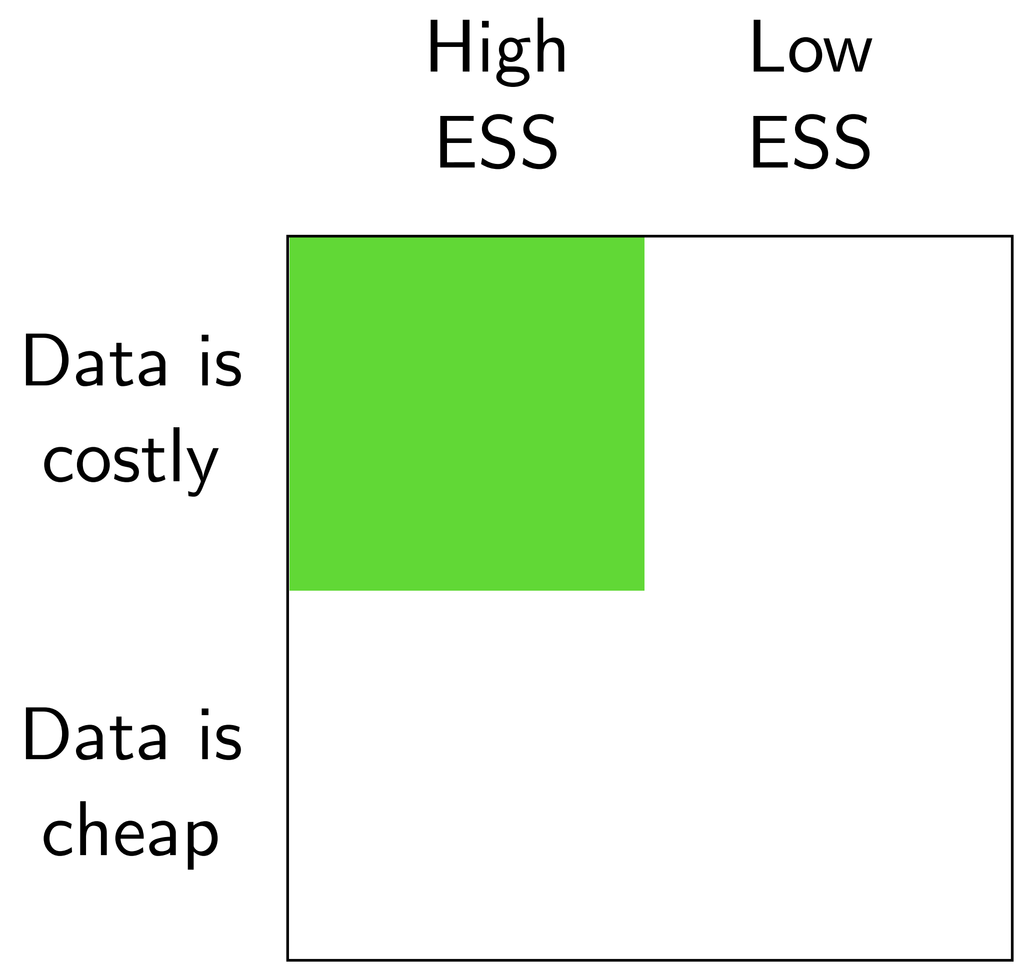
- Researchers/firms will increasingly use LLMs to predict economic behaviors and outcomes
- We need to understand when this is a good practice and when it is not
- We provide a measure to help with this evaluation
- The measure can help researchers understand and communicate the capability (or lack thereof) of LLMs across different domains

Conclusion

1. When is ESS high?

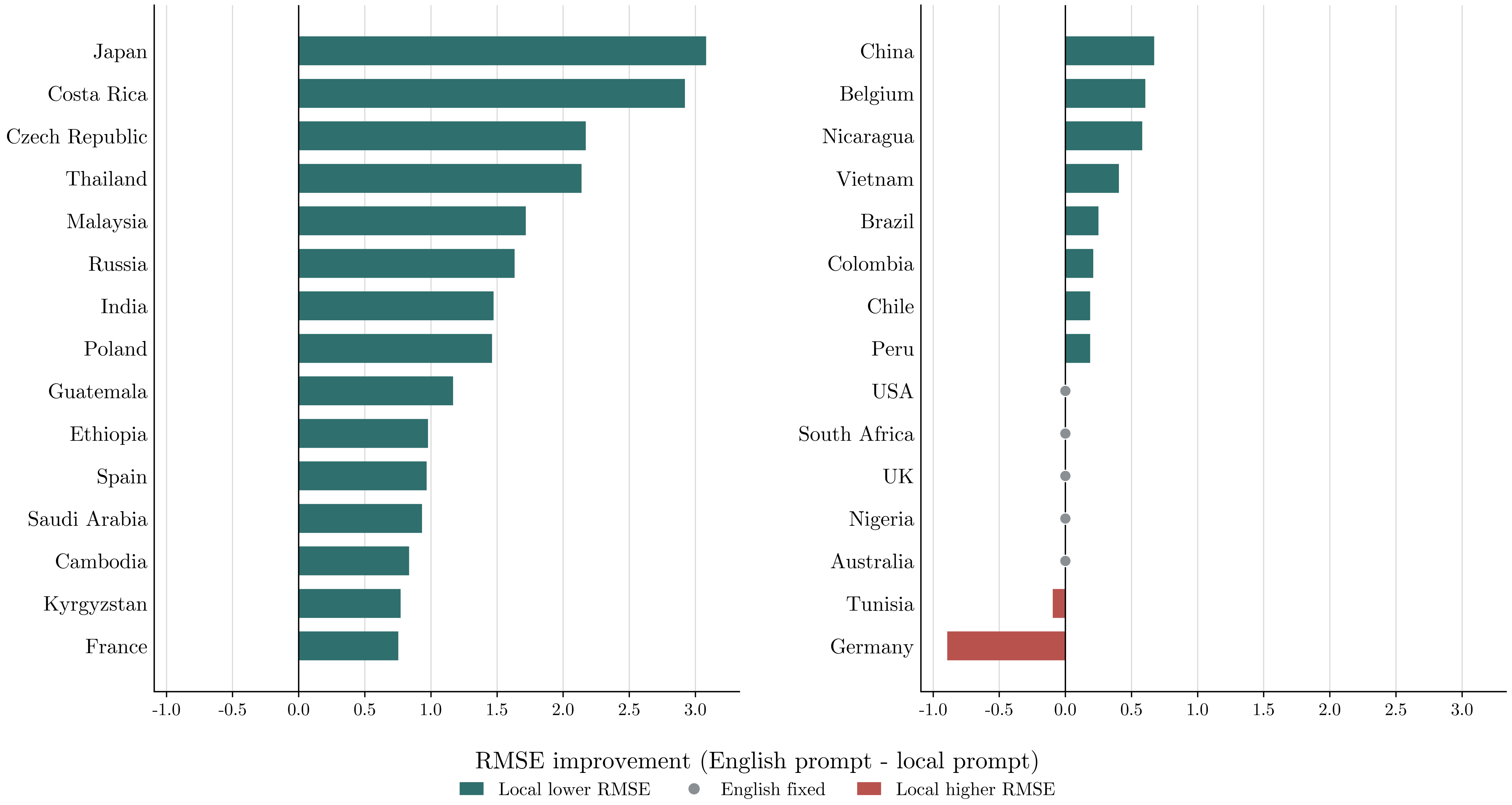


2. Where do LLMs have most value?



Change in LLM Error by Country

When is prompting in the local language most useful?



Variations on the LLM Prediction Rule

Alternative analyses:

- Use **DeepSeek** to make predictions

	GPT-5.4		DeepSeek	
Method	Median ESS	Crossings	Median ESS	Crossings
CPT	1	30/30	1	30/30
EU	4	29/30	2.5	30/30

LLM value **degrades**
(lower ESS)

Variations on the LLM Prediction Rule

Alternative analyses:

- Use **DeepSeek** to make predictions
- Ask the prompt **in Chinese**

	English prompt		Chinese prompt	
Method	Median ESS	Crossings	Median ESS	Crossings
CPT	1	30/30	6	27/30
EU	4	29/30	9	19/30

LLM value **improves**
(higher ESS)

Variations on the LLM Prediction Rule

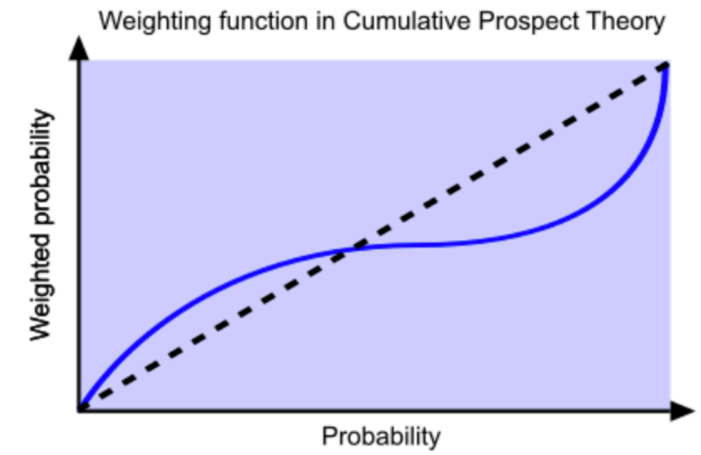
Alternative analyses:

- Use **DeepSeek** to make predictions
- Ask the prompt **in Chinese**
- Ask the prompt in **the primary language** of the country

	English prompt		Local prompt	
Method	Median ESS	Crossings	Median ESS	Crossings
CPT	1	30/30	5	27/30
EU	4	29/30	9.5	22/30

LLM value **improves**
(higher ESS)

Variations on the LLM Prediction Rule



Rough intuition:

- GPT 5.4 with English prompts is "too behavioral"

			Median fitted values	
Rank	Model / prompt	Distance from expected value	γ	δ
EU	Risk-neutral EU maximizer	0.000	1.000	1.000
1	GPT 5.4, Chinese prompt	1.119	0.865	0.889
2	GPT 5.4, Local language	1.393	0.838	0.875
3	GPT 5.4, English prompt	2.112	0.838	0.776
4	Deepseek, English prompt	2.931	0.600	0.713

probability distortion parameters